Worksheet 5: Fruit and Seed Structure

NCERT Class 11 Biology — Chapter 5: Morphology of Flowering Plants

Student Name: ____________________________	Roll No.: ______________	Section: _________

Section A: Multiple Choice Questions (Select the single best answer)

1. The fruit of Mango and Coconut is anatomically classified as a: [_____]

A) Berry
B) Capsule
C) Drupe
D) Legume

2. In a Coconut fruit, the edible coconut water/meat and the fibrous husk represent: [_____]

A) Endocarp and Epicarp respectively
B) Endosperm and Mesocarp respectively
C) Mesocarp and Seed coat respectively
D) Epicarp and Cotyledon respectively

3. Which of the following is an exception to the general rule that dicot seeds are non-endospermic? [_____]

A) Gram
B) Pea
C) Bean
D) Castor

4. The single, large, shield-shaped cotyledon in a monocot grain (such as Maize) is called: [_____]

A) Scutellum
B) Coleoptile
C) Coleorhiza
D) Aleurone layer

Section B: Fill in the Blanks & Exceptions

5. Fruits developed without the process of fertilization are called ________________________ fruits (e.g., Banana).

6. In monocot seeds, the plumule is enclosed in a protective sheath called ___________________, while the radicle is enclosed in ___________________.

7. Though most monocotyledonous seeds are endospermic, ________________________ are an important non-endospermic monocot exception.

Section C: Dicot vs. Monocot Seed Comparative Table

Structural Feature	Dicotyledonous Seed (e.g., Gram/Pea)	Monocotyledonous Seed (e.g., Maize)
Seed Coat Layers	Two distinct layers: outer testa & inner tegmen	____
Cotyledons	Two thick, fleshy cotyledons (food storage)	____
Endosperm Nature	Mostly non-endospermic (Exception: Castor)	____
Embryonic Sheaths	Plumule and radicle are naked	____

Specialized Layers	None	____

Section D: Short Answer & Analytical Questions

8. Describe the three distinct layers of the pericarp in a Mango drupe. Which layer is edible?

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____
____

9. What is the structural position and biological function of the aleurone layer in monocot grains?

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Lecturer Signature: ____________________________